

Ravi Sahu

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Summary

AI Engineer specializing in end-to-end LLM orchestration, agentic AI systems, and RAG architectures via LangChain and LangGraph. Delivered 3 fully deployed production AI systems in 2026, including a live multi-agent research platform on Hugging Face Spaces, with end-to-end MLOps and CI/CD practices.

Education

SSIPMT (Shri Shankaracharya Institute of Professional Management & Technology) **Raipur, C.G.**
B.Tech in Computer Science Engineering | CPI: 7.0/10 *2021 – 2025*

Experience

Intern Developer **Jul 2024 – Aug 2024**
Thought Applied Creations *Raipur, India*

- Improved page load performance by ~30% by debugging bottlenecks in backend and frontend components of client-facing web applications.
- Built and integrated 5+ REST API endpoints and UI components with hands-on full-stack development workflows.

Skills

Languages: Python, C++, JavaScript, SQL
AI/ML: LangChain, LangGraph, LlamaIndex, Hugging Face, RAG, NLP, Vector Embeddings, YOLOv8, OpenCV
Backend: FastAPI, REST APIs, Docker, Kubernetes (Basics), MLOps, CI/CD
Databases: ChromaDB, PostgreSQL, Pandas, NumPy, PyMuPDF
Frontend & Tools: React (Basics), Streamlit, Plotly, Git, GitHub
APIs: Groq API, OpenAI API, Anthropic Claude API

Projects

NEXUS — Multi-Agent AI Research & Business Intelligence System [GitHub](#) | [Live Demo](#)
Python, LangGraph, Groq LLaMA 3.3 70B, ChromaDB, Tavily, Plotly, Streamlit, Docker

- Built a 4-agent autonomous pipeline (Research, Analyst, Writer, Critic) via LangGraph state machine — reducing manual research time by 80%+.
- Implemented self-critique and auto-revision loop — reducing hallucination by ~40% across 3+ revision cycles.
- Integrated Tavily real-time web search, ChromaDB vector memory, and Plotly BI dashboards. Deployed live on Hugging Face Spaces via Docker with CI/CD.

Local RAG + AI Agent Platform

Mistral 7B, LangChain, ChromaDB, FastAPI, React, NLP, Vector Embeddings

- Built a fully local, privacy-first RAG pipeline — achieving 100% elimination of external API dependency via 4-bit quantized Mistral 7B, reducing inference cost to \$0.
- Designed autonomous NLP agents for search, summarization, and multi-step task execution with tool use and vector memory.
- FastAPI backend supporting 10+ concurrent sessions paired with a React dashboard for document management.

AI-Powered DDR Report Generator

[GitHub](#)

Python, Groq API, LLaMA 3.1, PyMuPDF, python-docx, REST APIs

- Designed an LLM pipeline merging 2 PDF sources into structured diagnostic reports with zero manual intervention — cutting generation time by ~90%.

- Engineered structured JSON prompts to enforce strict output format and prevent hallucination — achieving near-zero formatting errors.
- Automated Word document generation with severity-coded tables, area-wise observations, and image placement via python-docx.